



Anti-biofouling properties of amphiphilic phosphorylcholine polymer films[☆]

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ABSTRACT

Surfaces of amphiphilic phosphorylcholine polymer (PC1036) prepared by spin-coating were characterized by spectroscopic ellipsometry, water contact angle and atomic force microscopy. The antifouling properties of the PC1036 films to marine benthic diatom *Nitzschia closterium* MMDL533 were also investigated. The results showed that the dry PC1036 film promoted the adhesion of *N. closterium* MMDL533 because the hydrophobic lauryl groups were present in the film surface. The 2 h-swelled PC1036 films had excellent anti-fouling properties with extremely low attachment densities and retention densities no matter what the annealing temperature was. The thickness of the coated films lower than 147 Å had a profound effect on the film anti-fouling properties. Otherwise, when the film thickness was higher than that value, there was no more improvement of diatom cell reduction observed. The annealing temperature had only a little effect on the film resistant to diatom adhesion, which might be attributed to two factors including the PC group packing densities in the outer PC layer and the equilibrated water volume fraction in the 2 h-swelled PC1036 films.

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1. Introduction

The accumulation and growth of marine organisms on underwater artificial surfaces bring gross economic losses to human being [1]. The toxic anti-fouling coatings could prevent adhesion of fouling organisms efficiently and had been used to protect solid surfaces for a long time [2]. However, the leaching of toxin from such coatings into seawater had caused serious irreversible destruction to ocean ecosystem [2]. Technologists are in search of and the industry is in demand of efficient yet inexpensive non-toxin anti-fouling strategies to replace the toxic ones.

Marine cells often attach to solid surfaces via the adhesive extracellular polymeric substances (EPS) [3]. The properties of EPS determine the adhesion behavior of the certain fouling cells. Since the macromolecules including proteins, glycoproteins and heteropolysaccharides in EPS are likely to be amphiphilic, they should undergo conformational changes to adapt the solid surfaces with different physicochemical properties to optimize the adhesion

process [3,4]. These environmental regulation mechanisms make marine cells have the ability to adhere to all kinds of solid surfaces which consequently cause much more difficulties for anti-fouling research. Besides this, the properties of marine cell settlement and adhesion vary from one species to another. For example, the hydrophobic poly(dimethyl siloxane) (PDMS) strongly promoted the adhesion of macroalga *Ulva linza* spores and sporelings, and their adhesion strengths were significantly weaker on PDMS, as compared with the hydrophilic glass substrates [4]. In contrast, though PDMS could also promote the adhesion of diatoms, the adhesion is so strong that the adhered diatoms are not easily released from PDMS-based fouling-release coatings [4]. Therefore, more environmental and effective anti-fouling strategy should be developed in future.

It had been reported that the hydrophilic and hydrophobic domains of amphiphilic polymer networks could undergo phase-segregation which resulted in the reduction of protein adsorption or a higher release or *Ulva* sporelings [4]. The phosphorylcholine (PC) copolymers containing synchronously hydrophilic PC block and hydrophobic lauryl block produce such kind of amphiphilic networks. In fact, there were several articles had reported that PC copolymer film could resist protein adsorption and cell adhesion effectively [5,6]. Therefore, it is reasonable to assume that PC copolymer film can also resist the adhesion of marine organisms based on the same principle. Since the PC copolymers mimic the natural PC lipid molecules in the cell wall, the polymers are environmentally friendly materials and inherently resist biofoul-

[☆] In this paper, the new antifouling material amphiphilic phosphorylcholine (PC) polymer had been used to reduce the marine fouling organism adhesion for the first time. Both the attachment densities and adhesion strength of the benthic marine diatom were extremely low which indicated that the introduction of the PC groups in polymer materials should be a potential marine antifouling strategy.

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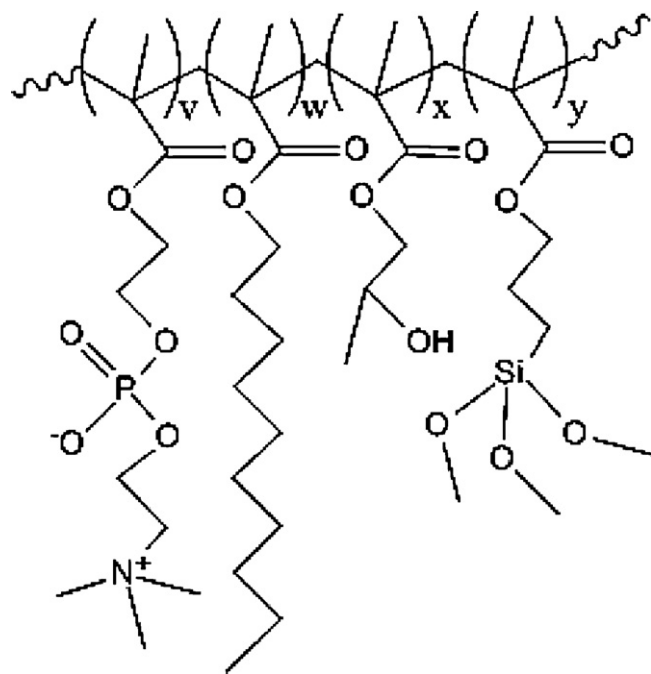


Fig. 1. Molecular structure of PC1036 copolymers.

ing by proteins [7]. In this report, we investigated the ability of PC1036 copolymer to resist diatom adhesion as well as the release of attached cells after exposure to wall shear stress.

2. Materials and methods

2.1. PC copolymers

Phosphorylcholine (PC)-incorporated copolymer based on 2-methacryloyloxyethyl phosphorylcholine (MPC) were obtained from Biocompatibles U.K., Ltd. The general molecular structure of the copolymers was shown in Figs. 1–4. The silyl cross-linking agent in the form of 3-trimethoxysilylpropyl methacrylate (TMSM) was fixed at 5%. Incorporation of 15% 2-hydroxypropyl methacrylate (HMP) was necessary to adjust the extent of water intake into the hydrogel material. The hydrophobic lauryl methacrylate (LM) was 51% and the hydrophilic MPC was 29%.

2.2. Preparation of PC polymer films

All silicon wafer surfaces used in the present work were cut from silicon (1 1 1) wafers (Compact Technology Ltd., UK). The thickness of its thin native oxide layer was determined to be about 13 Å using ellipsometry. All glass samples used here were cut from normal microscopic slides with a diameter of 25 mm. Prior to surface coat-

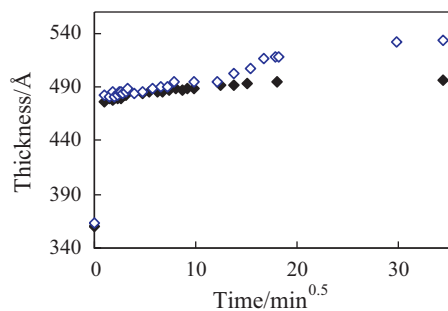


Fig. 2. Change of thickness with time for the thin PC1036 films annealed at 150 °C during swelling in water (♦) and ASW (◇).

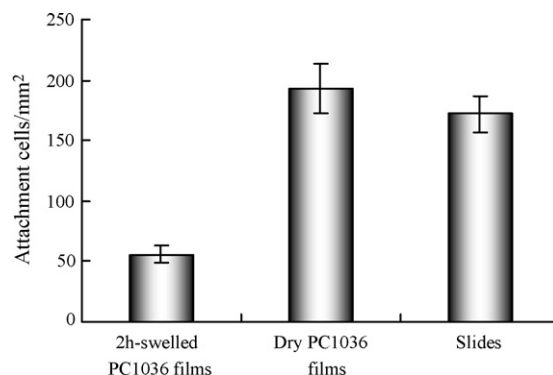


Fig. 3. The influence of the swelling process on the attachment of *Nitzschia closterium* MMDL533 on PC1036 films with the thickness of 147 Å annealed at 150 °C.

ing, both the 25 mm slides and the silica wafers were cleaned with 5% Decon90 solution by weight (Decon Laboratory, UK), followed by copious rinsing with Millipore water (resistivity >18 MΩ/cm). The surfaces of silica were proved very hydrophilic and are ready for spin coating. Afterwards, all cleaned glass slides were transferred into a Piranha solution (7:3 concentrated H₂SO₄/30%H₂O₂) for at least 1 h at 100 °C to remove traces of adsorbed organic materials. The glass samples were then rinsed judiciously with deionized water, followed by Millipore water. After thorough washing using Millipore water, the contact angles of the slides were found to be less than 5°, indicating that the surfaces were hydroxylated. The glass slides were then removed and dried, ready for either diatom adhesion studies on bare slides or spin coating experiments.

All engineered surfaces were prepared by spin coating. The PC1036 polymers were dissolved in absolute ethanol under stirring at room temperature overnight with a concentration of 1.2 wt.%. Polymers were then filtered with 0.22 μm filters to remove possible insoluble particles. The PC films were prepared by adding a sufficient quantity of precursor solution to completely cover

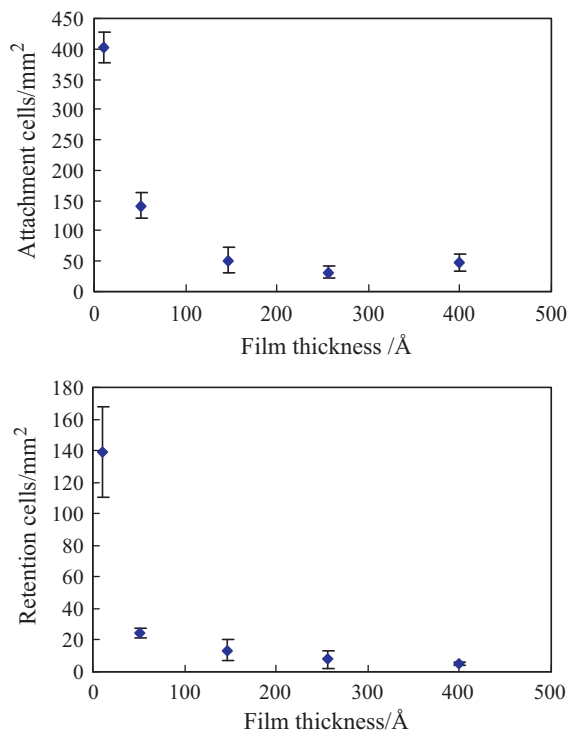


Fig. 4. Attachment (upper) and retention (lower) after exposure to 8.5 Pa wall shear stress of *Nitzschia closterium* MMDL533 on PC1036 films annealed at 150 °C. Each point represents the mean value from 30 counts from 3 replicate slides.

the stationary substrate surface, typically 90 μL , and the sample rotation was conducted immediately at 5400 rpm for 60 s, using a spin-up rate of 240 rpm. The coated films were then dried in air for at least 30 min at room temperature before being thermally annealed at a certain temperature for 3 h under vacuum. The annealing temperature was 150 °C unless the special coating conditions were illustrated. The samples were then left to cool to room temperature under vacuum and were stored in vacuum desiccators for subsequent use.

Three kinds of wafer including 1.9 cm $\times$ 1.4 cm silicon for swelling experiments, 2.5 cm $\times$ 2.5 cm silicon for thickness determination and 2.5 cm diameter slides for biofouling assays were used here. The film thickness of slides was not determined directly but was regarded as the same value with the 2.5 cm $\times$ 2.5 cm silicon wafer coated at the same conditions.

2.3. Surface characterization

The thickness and optical constants of polymer films coated on a silicon oxide substrate were measured with variable angle spectroscopic ellipsometry (SE) (Jobin-Yvon UVISSEL, France). The coated wafer was positioned in the central area and the beam was directed to the sample surface at an incidence angle of 74.5° with respect to the surface normal. The ellipsometric measurement was carried out for the determination of the thickness of the films in open air. Prior to the SE measurements, all films were dried in vacuum for 2 h under ambient temperature. SE measures the change of the state of polarization of light reflected at the interfaces. The change of polarization is usually expressed as two ellipsometric angles, Ψ and Δ , and is dependent on the thickness of any surface layer and its optical constants (refractive indices). Simultaneous measurements of Ψ and Δ over a range of wavelengths provide information about the polymer film coated in abstract space, and the usual route to convert such information into layer modeling to the pairs of the measured Ψ and Δ . For all the films studied in this work, uniform layer model was found to be adequate for describing them. This was evident from the good agreement between the measured data and calculated fits. To improve the consistency of the data analysis, the refractive indices for the thin polymer films were taken to the same as obtained for the pure polymer to avoid any uncertainty caused by the correlation between thickness and refractive index when the films were very thin. Such treatment should not affect the estimation of the amount of polymer coated in a given surface area. Ψ and Δ were recorded simultaneously over the wavelengths between 350 and 700 nm. The time required for each scan over this wavelength range was determined by the number of pairs of Ψ and Δ counted. It typically took about 40 s to complete one scan involving about 30 pairs of Ψ and Δ measurements. The experimental data were analyzed using DeltaPsi 2 software (Jobin-Yvon) [7]. For each coated sample with the dimension of 2.5 cm $\times$ 2.5 cm, measurements of Ψ and Δ at the air/solid interface were made at five different spots. Usually, one spot were taken at the center of the square, and the other four spots were taken 0.5 cm away from each corner. The thicknesses from all the five independent measurements were recorded and the mean value was taken to be equal to the sum of the total thicknesses divided by the number of runs.

The swelling process of PC1036 film was monitored by determination of film thickness by SE according to the reported methods [8]. Briefly, the coated silicon wafer was positioned in the central area of a specially designed cell and was held by a sample holder. The ellipsometric measurement was usually started with alignment at the air/solid interface first, followed by the determination of the thickness and refractive indices of the dry film in open air. After the introduction of water into the cell, the subsequent autoscan were repeated at the solid/solution interface. The ellipsometric data were analyzed to obtain the best-fit film thickness and refractive indices

using a Levenberg–Marquardt algorithm according to Tang et al. [8].

The contact angles (CA) of the obtained samples were determined by the sessile drop technique using an Automatic Contact Angle Meter SL200B (Solon Tech. Co., Ltd., Shanghai) in the laboratory atmosphere. The probe liquid was water. The CA results were averaged over at least five parallel measurements at different positions on the substratum surface.

Atomic Force Microscope (AFM) experiments were conducted with a commercial Nanoscope IVa MultiMode AFM (Digital Instruments, Santa Barbara, CA) equipped with an AS-12 Scanner. To minimize the force exerted from the scanning tip on the PC films, the AFM was operated in tapping mode at room temperature. All the measurements were performed in air for higher lateral image resolution, and a TESP silicon probe (Veeco, Santa Barbara, CA) with a nominal spring constant of 42 N/m and a typical frequency 300 kHz was used in all image collections. The obtained PC1036 films were subjected to AFM imaging directly. Images were collected at a scan rate of 1 Hz and at a scan angle of 0°. All images are flattened using a first-order line fit (the flatten function in AFM software) to correct for piezo-derived differences between scan lines. Measurements were made at least three times in each case and representative images are shown here.

2.4. Biofouling assays

Slides were incubated in sterile artificial seawater (ASW) for at least 2 h at about 20 °C unless the special statement was given prior to the start of every experiment to achieve equilibration with seawater.

Nitzschia closterium MMDL533 was obtained from School of Life Science, Xiamen university, China. The strain was isolated from Chinese offshore near Xiamen, and is a normal marine benthic diatom in intertidal zone. *N. closterium* MMDL533 was cultured in F/2 medium and diatom suspensions were prepared as described in Li et al. [9]. Twenty five milliliters of the cell suspension was carefully pipetted into a 9-cm-diameter Petri-dish containing 6 glass slides (25 mm diameter) on the bottom. After 2.5 h settlement in the light, the glass slides were gently washed in ASW to remove cells that had not properly attached. For static settlement assays, the slides were fixed in 2% (v/v) glutaraldehyde in ASW for 10 min, followed by washing sequentially in distilled water before drying. Attachment densities were obtained by counting the adhered cells using a digital microscope (Leica DMI 3000B) combined with image software (Leica Qwin V3 imaging workstation). Counts were made for 10 fields of view on each of 3 replicate slides. Fields of view were taken at 1-cm diameter area of the samples to avoid the effect of manipulation during experiments. For detachment studies, slides were incubated for a further 2 h before exposure to shear stresses in flow apparatus. The release experiments were conducted according to our reported methods [9]. Briefly, unfixed slides after the initial settlement period were exposed to wall shear stress of 8.5 Pa in a flow apparatus for 2 min. The flow apparatus was constructed according to the description of Schultz et al. [10] with some modification [9]. The retention densities were obtained by counting the number of diatoms still attached to slides after exposure to the flow.

The aggregation behavior of the attached diatoms had also been recorded. After fixation, cells adhered to the samples were viewed under objective lenses (20 $\times$ or 10 $\times$) using the digital microscope. Digital images were captured by the imaging software directly to observe the aggregation behavior of the attached cells. Data are presented below for the mean numbers of diatom cells that adhered $\pm$ 95% confidence limits ($x = 20$) and also for the percentages of total diatom cells that were present in groups. The percentages of cells found in groups were calculated by combing group sizes as follows: 1, 2–5, 6–9, 10–15, 16–19, and ≥ 20 .

Table 1

Determination of the thickness of PC1036 polymer films by spectroscopic ellipsometry under different coating conditions.

PC polymer	Concentration (wt.%)	Rotation speed (rpm)	Precursor solution quantity (μL)	Thickness ($\text{\AA}$)
PC1036	0.01	5400	90	11.3
PC1036	0.1	5400	90	50.5
PC1036	0.4	5400	90	147.4
PC1036	0.8	5400	90	256.6
PC1036	1.2	5400	90	400.2

3. Results and discussion

3.1. PC polymer film formation

PC1036 films were coated onto optically flat silicon wafers by spin coating to facilitate the determination of film thickness by SE. The Ψ and Δ of obtained films were measured by SE at the air/PC polymer-coated interface. The measurements were made at an incidence angle of 74.5° with respect to the surface normal and at an ambient temperature of 20°C . The wavelength dependence of the refractive index of the polymer film was taken as $n = A + B \times 10^4/\lambda^2$ ($A = 1.475$, and $B = 0.48$) following the Cauchy equation [7]. The precursor solution quantity was adjusted to obtain copolymer films with different thicknesses (Table 1). In our experiments, PC1036 films were prepared with the thickness of 11–400 $\text{\AA}$.

3.2. Swelling of PC1036 films

Swelling in water could change the surface structure and physiochemical properties of polymer films as a result of water penetration in the films. The PC copolymer films had been found more hydrophilic upon immersion in water for a certain time which could consequently influence the protein adsorption as well as cell adhesion. Therefore, we had used SE to study the swelling process of PC1036 films in water and ASW respectively. The results showed that PC1036 films swelled according to a two-step pattern as the swelling behavior of the PC100B films [8] both in water and ASW. Briefly, the swelling process composed of two stages, an initial diffusion-controlled fast swelling followed by a subsequent slower process controlled by the relaxation of polymer fragments. The increase of film thickness in ASW was higher than in water with the same immersion time. For example, when the incubation time was 323 min, the thickness of PC1036 film changed from 360 $\text{\AA}$ to 495 $\text{\AA}$ in water and from 364 $\text{\AA}$ to 518 $\text{\AA}$ in ASW.

3.3. Marine anti-fouling properties

3.3.1. The influence of film swelling on its anti-fouling properties

The effect of PC1036 film swelling on the diatom attachment and retention had been studied here. The so-called “dry film” meant those films that were directly used for the adhesion experiments without any swelling process. The results showed that the dry PC1036 films did not resist diatom cell adhesion as compared to the hydrophilic hydroxylated slides. However, both the attachment and retention densities on the 2 h-swelled PC1036 films were dramatically low. The hydrophobicity of dry and 2 h-swelled PC1036 films had been studied by water contact angle measurement. The results showed that the water contact angle of PC1036 film decreased from 95° to 53° after 2 h swelling, indicating that the hydrophobic lauryl groups were present in the dry PC1036 film. Since the hydrophobic alkyl groups could promote *N. closterium* MMDL533 adhesion [9], it was well explained that the dry PC1036 film could not resist diatom attachment. Since the fast step of PC1036 film swelling occurred at the first minute after immersion in water, we could speculate that even the “dry” PC1036

films had also swelled to a certain degree when contacted with most attached diatom cells. Compared with the extremely low cell attachment densities on 2 h-swelled PC1036 films, dramatically more cells attached on “dry” films might indicate that the second stage of swelling played a more important role in the film anti-fouling properties.

When the PC films were immersed into water or ASW, water penetrated into the films and the film surface structure was changed immediately as showed by SE data in Section 3.2. It is well known that amphiphilic polymer carry surface reconstruction upon immersion into water to low system energy. Krishnan et al. [4] found that the comb-like amphiphilic block copolymers containing hydrophilic block PEG and hydrophobic fluoroalkyl block resisted *Ulva* zoospores, *Ulva* sporelings and diatom *Navicula* cells effectively compared with PDMS. The proposed anti-fouling mechanism was the occurrence of amphiphilic polymer surface reconstruction in water which made the hydrophilic PEG block expose to water and the hydrophobic fluoroalkyl block fold into the film interior. In our case, swelling in water made the hydrophilic PC containing block expose to water and the hydrophobic lauryl containing block concealed into the film interior. In fact, a similar amphiphilic copolymer films containing dodecyl and phosphorylcholine side chains had been studied for the ability to reduce protein adsorption. It was found that the outer polymer surface should be covered by a layer of hydrophilic PC groups upon immersion in water [6] and the methacrylate back bone and the lauryl groups made a hydrophobic environment in film interior. Therefore it was reasonable to speculate that the occurrence of surface construction made the swelled PC1036 films resist to diatom adhesion. It was reported that the second stage of PC100B film swelling controlled by relaxation of polymer fragments approximately contributed to about 50% of the total water uptake [8]. Combined with the data with the dry PC1036 films, our results might indicate that the surface reconstruction occurred during the second stage of the PC1036 film swelling.

3.3.2. The influence of film thickness on diatom adhesion

It had been reported that PC polymer film thickness could influence its macromolecular loading capacity and resistance to protein adsorption, diatom adhesion which acts through the adhesive macromolecular could also vary according to the film thickness. Our results showed that diatom attachment density decreased sharply from 403 ± 25 cells/ mm^2 to 52 ± 21 cells/ mm^2 when the thickness of the PC1036 film increased from 11 $\text{\AA}$ to 147 $\text{\AA}$. When the thicknesses of the PC1036 film fell in the range of 147–400 $\text{\AA}$, the attachment densities were extremely low with a value range of 32–52 cells/ mm^2 .

PC1036 polymer reacts with the hydroxyl groups in slide surface through the methoxyl groups in the TMSM side chains. When the film thickness was 11 $\text{\AA}$ which is even thinner than the full extended length of the lauryl side chain of about 15 $\text{\AA}$ [11], the interior of ultra thin film had not enough space to accommodate the hydrophobic lauryl groups. The AFM results in Figs. 5–7 apparently show the different morphology varying film thickness. The 11 $\text{\AA}$ film surface was typically rougher than the other films. There was almost no smooth area on the 11 $\text{\AA}$ film surface which indicated that the polymer network had not been formed. The lauryl groups in the thin PC1036 film surfaces promoted the adhesion of *N. closterium* MMDL533 cells. Along with the film thickness increased, the methoxyl groups reacted with the hydroxyl groups in the PC1036 side chains which induced the formation of complicated polymer networks. This had been proved by the AFM images which showed that the thicker surfaces became smoother. When the film thickness was higher than 147 $\text{\AA}$, the surface morphology exerted little different with the roughness value with a range of 3.0–4.2 $\text{\AA}$ in an area of $500 \text{ nm} \times 500 \text{ nm}$ calculated by the AFM image software. This roughness range was apparently lower than the 9 $\text{\AA}$ of

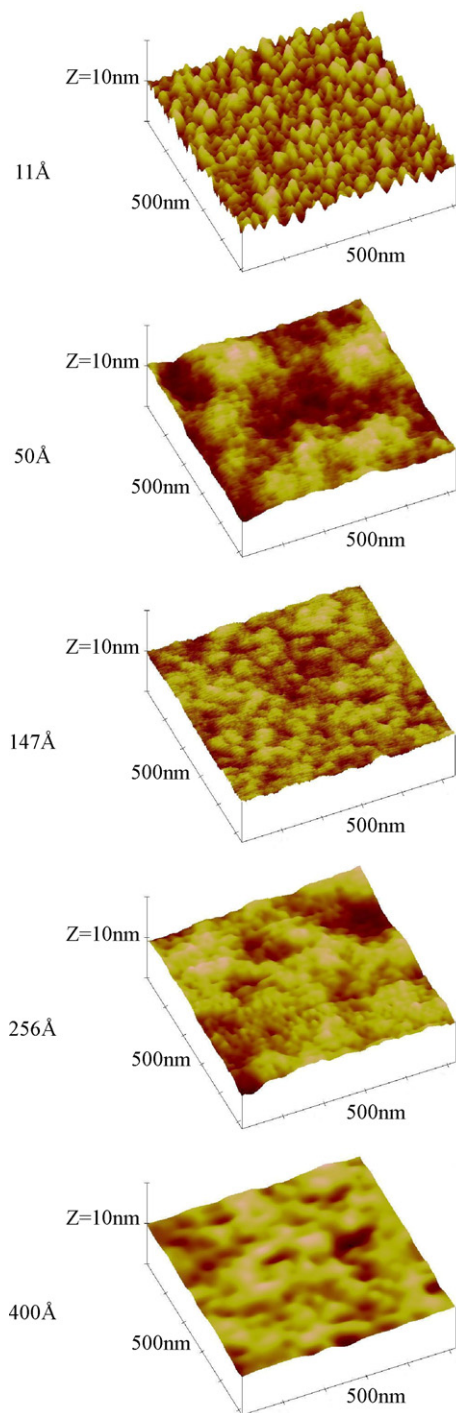


Fig. 5. The AFM images of the PC1036 films. Images were captured by tapping mode with a scan rate of 1 Hz.

11 Å film and 6 Å of 50 Å film calculated in the same way. The formation of the polymer network in thicker films made the surface properties very similar. For thicker films, there was enough space in the film interior to carry structure reconstruction. As a result, more lauryl groups were concealed into the film interior and more PC groups were exposed to water. The PC groups in the thicker film surfaces packed to form a hydrophilic layer out the film. Since the anti-fouling properties of polymer films were determined by the film surface morphology and physiochemical properties but not the whole film properties [6]. Therefore, the PC layer reduced the attachment densities on thicker films. The diatom attachment densities remained at a stable level when the thickness of PC1036

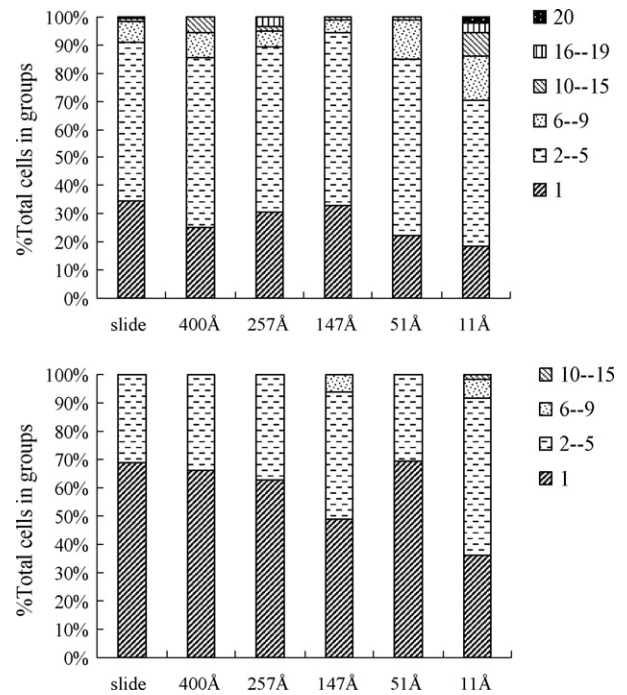


Fig. 6. Percentages of total *Nitzschia closterium* MMDL533 after 2.5 h settlement (upper) and after exposure to 8.5 Pa of 2 min wall shear stress (lower) in groups of various versus the thickness for PC1036 copolymer.

film got higher than 147 Å might be due to their similar surface properties.

The retention of the attached organism cells on a certain film after exposure to wall shear stress had also been used to judge the film anti-fouling capacity. Though the initial attachment densities

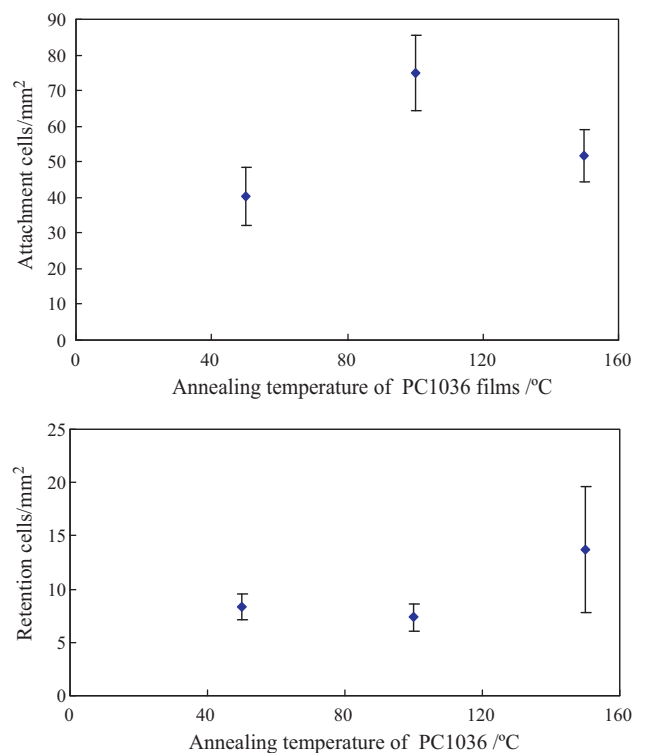


Fig. 7. Attachment (upper) and retention (lower) after exposure to 8.5 Pa wall shear stress of *Nitzschia closterium* MMDL533 on PC1036 films with the thickness of 147 Å. Each point represents the mean value from 30 counts from 3 replicate slides.

were dramatically different in our experiments, the environmental factors including temperature, solutions, etc. were same for the each kind of films. Therefore, we focused on the retention densities of the attached diatom cells while concerning the concept of the resistance of attached cells to wall shear stress. Our results showed that after exposure to 8.5 Pa wall shear stress of 2 min, the retention of attached diatom cells on PC1036 films showed the similar trend compared with the attachment densities. When the film thickness was higher than 147 Å, the attached diatoms were almost removed with a value range of 5–14 cells/mm². The retention of attached diatom cells after exposure to wall shear stress was $5 \pm 1 \text{ mm}^{-2}$ at 400 Å, which was much lower than the value of $129 \pm 28 \text{ mm}^{-2}$ at 11 Å. Since on thinner films, there were more defects and cavities on the outer PC layer which consequently provided more adhesion sites for diatoms. Surface adsorption site made the diatom adhesion stronger which induced the higher retention densities.

The morphology of the attached diatom cells on PC1036 films was dramatically different along with the change of film thickness. There was more and bigger cell aggregation occurrence on the thinner films. We had found that hydrophobic CH₃-SAMs could promote the co-adhesion of diatom cells which consequently made the more uneven distribution of the attached diatom cells in our previously work [9]. Here, the lauryl groups in the PC1036 films might promote the gregarious adhesion of diatom. On the thicker PC1036 films, the distribution of the attached cells was most in single cell or small aggregation which might be due to the absence of the hydrophobic lauryl groups on the outmost surface of the polymer films. After exposure to wall shear stress, the cell aggregations bigger than 10-cells were totally removed. Compared with the single cell or small aggregations, bigger cell groups seemed more easily to be removed. This investigation was consistent with our previous work [9].

3.3.3. The influence of film annealing temperature on diatom adhesion

In order to remove remaining solvents trapped in the layer and relieve any strain produced during the film formation, the PC polymer films were annealed by heating the polymer to a certain temperature in vacuum. It had been reported that annealing temperature exerted influence on the resistant ability of the PC100A films to lysozyme [6]. Therefore, we investigated the effect annealing temperature of PC1036 films on the diatom adhesion behaviors. The results showed that the PC1036 films with the thickness of 147 Å did have excellent anti-fouling properties with low diatom attachment densities and low diatom retention densities after exposure to 8.5 Pa wall shear stress of 2 min regardless of the annealing temperature. However, the relationship between film annealing temperature and film resistant capacity to diatom adhesion was not linear. Both the attachment densities and retention densities were highest when the annealing temperature was 100 °C. Higher annealing temperature could improve the uniformity of the PC polymer film, and thus improve the packing densities of the outer PC layer, leading to a further reduction in the protein deposition [5]. In our case, the high ionic strength of the ASW media used here decreased the electrostatic force between adjacent PC groups which drive the formation of the outer PC layer. However, the adhesion densities of the PC1036 films were extremely low which indicated that the outer PC layer had formed even in the high ionic strength environment. When the annealing temperature increase, the well-packed PC layer on the outer surface of the PC1036 films with higher annealing temperature could reduced macromolecule adsorption which in turn reduced diatom adhesion. Since the PC100A films annealed at lower temperature had gotten through faster relaxation which lead to higher equilibrated water volume fraction [8], the 2 h-swelled PC1036 films

used in our experiments might experiment the same swelling process. The equilibrated water volume fraction had been proposed to affect the protein reduction ability of polymer films. According to Murphy et al. [6], the nonrandom water associated with the PC groups created a strong steric barrier prohibiting protein deposition. Thus, the PC1036 films annealed at lower temperature with higher equilibrated water volume fraction could result in more heavy hydration and stronger steric barrier, which consequently appeared more resistant to diatom adhesion. In conclusion, it might be the combination of the PC packing densities and the equilibrated water volume fraction simultaneously determined the non linear relationship between the annealing temperature and anti-fouling properties of the PC1036 films.

4. Conclusion

The dry PC1036 film promoted adhesion of *N. closterium* MMDL533 because the hydrophobic lauryl groups were present in the film surface. The 2 h-swelled PC1036 films had excellent anti-fouling properties with extremely low attachment densities and retention densities no matter what the annealing temperature was. The thickness of the coated films lower than 147 Å had a profound effect on the film anti-fouling properties. Otherwise, when the film thickness was higher than that value, there was no more improvement of diatom cell reduction observed. The annealing temperature had only a little effect on the film resistant to diatom adhesion, which was attributed to two factors including the PC group packing densities in the outer PC layer and the equilibrated water volume fraction in the 2 h-swelled PC1036 films.

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